BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Common Network Protocols

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is an IPv4 address used for and how is it usually written?

2. What is a router's main job?
   A) Store user passwords
   B) Print network documents
   C) Run spreadsheet formulas
   D) Link different networks

3. "The leftmost bit is always the least significant bit." Correct this. Explain why.

4. Tablets need to join a wired LAN. "A WAP changes domain names into IP addresses." Correct this. How does a WAP help the tablets?

Today's objectives

State purposes of commonly used network protocols. Compare secure web access and email protocols.

Key vocabulary - Read, cover, recall

Protocol
Rules for communication between devices.

HTTP
Hypertext Transfer Protocol, transfers web resources.

HTTPS
Secure HTTP with encrypted communication.

FTP
File Transfer Protocol, transfers files.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Protocols are rules

Communication protocols are agreed rules controlling how data is exchanged across a network.

Web and file protocols

HTTP transfers web pages; HTTPS uses encryption for secure web communication; FTP is used for file transfer.

Email protocols

SMTP sends email; POP downloads email to a client; IMAP accesses and synchronises server-based mail across devices.

TCP and IP

TCP helps ensure reliable, ordered delivery. IP supports addressing and routing so packets reach the intended destination.

I Do - Worked model

A teacher sends an email on a laptop, then reads received messages on a phone. Explain the roles of SMTP and IMAP. [4 marks]

- Sending and accessing messages are two separate jobs.
- SMTP is used to send outgoing email [1] from client towards email server [1].
- IMAP accesses mail stored on a server [1], allowing messages to be synchronised across devices [1].
- Name each protocol and link it to the specific email action.

We Do - Guided practice

1. A student visits a website requiring login details. Explain why HTTPS is preferable to HTTP. [4 marks]

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2. A student uses an email app on several devices. Explain the difference between POP and IMAP. [4 marks]

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3. Describe the different roles of TCP and IP in sending data. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the message.

```
protocol = "HTTPS"
if protocol == "HTTPS" then
print("Secure web protocol")
endif
```

You Do - Independent exam questions

Q2. State one function of SMTP. [1 mark]

Q3. A charity accepts donations on a public website. Explain how HTTPS differs from HTTP and why that matters when supporters enter payment information. [4 marks]

Q4. Describe the uses of FTP, SMTP, IMAP, TCP and IP. [6 marks]

Exit Ticket - True or False?

SMTP is mainly used to send emails.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots.

2. D - Link different networks

3. The MSB is the leftmost bit, weight 128; the LSB is the rightmost bit, weight 1.

4. A wireless access point connects wireless devices to a network. It has a different job from a switch or router.

I Do - Explain the model

Sending and accessing messages are two separate jobs.

SMTP is used to send outgoing email [1] from client towards email server [1].

IMAP accesses mail stored on a server [1], allowing messages to be synchronised across devices [1].

Name each protocol and link it to the specific email action.

We Do - Model answers

1. HTTPS carries web data [1] using encryption [1], making captured traffic unreadable without the key [1] and better protecting sensitive information during transfer [1].

2. POP retrieves/downloads mail to a client [1], often for local storage [1]. IMAP keeps/accesses mail on the server [1] and synchronises changes across multiple devices [1].

3. TCP helps ensure reliable delivery [1] in the correct order [1]; IP supplies addressing [1] and routing to the destination across networks [1].

Code First - Answer

```
protocol = "HTTPS"
if protocol == "HTTPS" then
print("Secure web protocol")
endif
```

Expected output: Secure web protocol

You Do - Self-mark

Q2: Sends email between clients/servers or mail servers [1].

Q3: HTTP is used to transfer web resources [1]. HTTPS adds encryption to web communication [1], so an interceptor without the appropriate key cannot read captured payment data [1], protecting sensitive information in transit [1].

Q4: FTP transfers files [1], SMTP sends mail [1], IMAP accesses/synchronises received mail [1], TCP checks reliable and in-order delivery [1], IP addresses packets [1] and routes them between networks [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.